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IoT Gateway Blueprint with Thread and Matter

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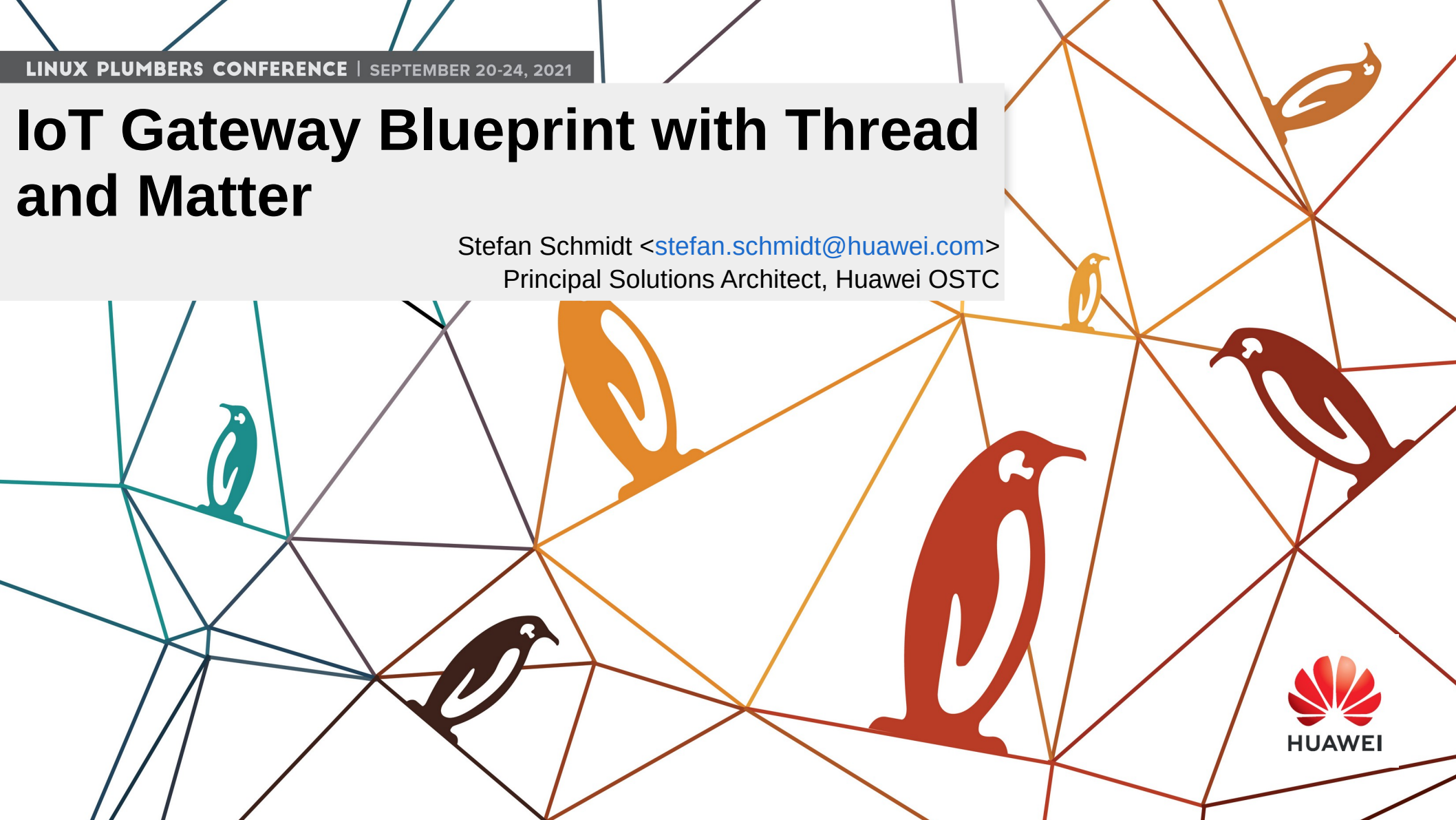


Name 3 things that come to your mind when hearing IoT Gateway.

Predicted Answers

- Connectivity: WiFi, Bluetooth, ZigBee, Z-Wave, ...
- One box for each system you use
- Not ***another*** one, please!
- Turn on and forget (to update)

Predicted Answers (primed)

- OpenThread
- Matter (former CHIP, Connected Home over IP)
- IPv6 as end-to-end solution all the way from backend to sensor
- Keep these devices ***updated***, please!

Agenda

- A discussion session, not a talk
- Ideas and implementations for an IoT gateway blueprint
- OpenThread and Matter open source projects
- IPv6 connectivity
- Zephyr nodes
- Assisted OTA service for nodes

Context

IoT Gateway Blueprint

- Partners interest in reducing R&D costs by not having one hub each
- Could the zoo of smart IoT hubs be reduced?
- Ignoring branding and vendor lock-in topics here, technology only!
- Brainstorming on current landscape and technologies available and/or upcoming

All Scenarios OS Gateway

- Based on All Scenarios OS (ASOS) with Yocto and Linux
- WiFi AP functionality, headless WiFi onboarding (maybe BLE, PWA)
- OpenThread Border Router service (Threadgroup Android app)
- Basic Web UI
- Sysotad is running for updates
- Able to host containerized services to enable device-specific services
- Matter protocol support

https://git.ostc-eu.org/OSTC/requirements/-/issues/13

(Brave) Future Prediction

- The usual bundle of IoT gadget and companion hub will slowly fade
- We will ***not*** end up with a single IoT gateway for all devices :-/
- Non-IP(v6) based IoT protocols will fade away
- Devices have a mandatory update mechanism (not a bold prediction as it gets backed up by laws)

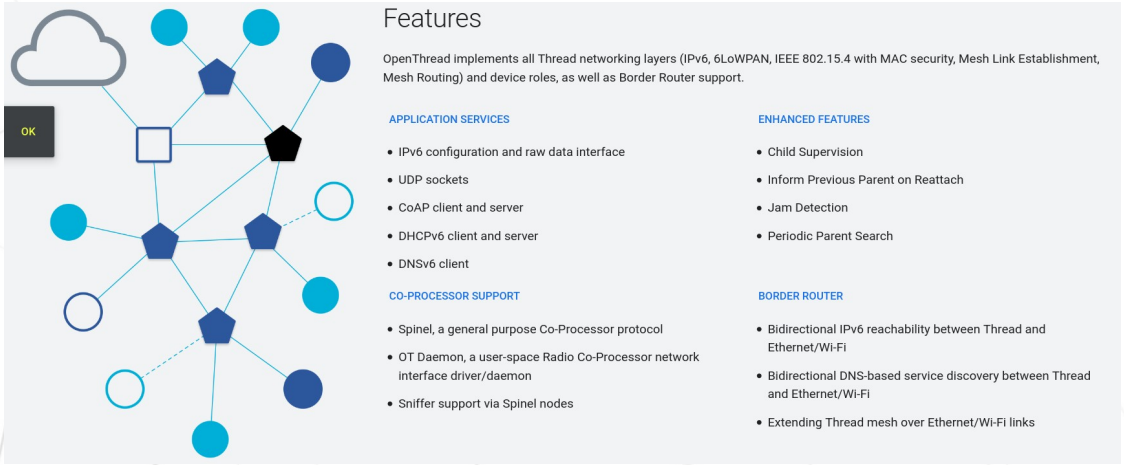
Discussion

Discussion Topics

- Aim for end-to-end solution without translating proxies on the hub e.g. IPv6 from device to device or cloud
- IPv6-only solutions (OpenThread, Matter)
- Native IPv6 in a home setting e.g. prefix delegation from ISP router
- NAT64 for IPv4-only transit networks (tayga, jool)
- Sandboxes for vendor specific code (e.g. backend connection, updates, etc) to avoid an extra hub for just software

OpenThread

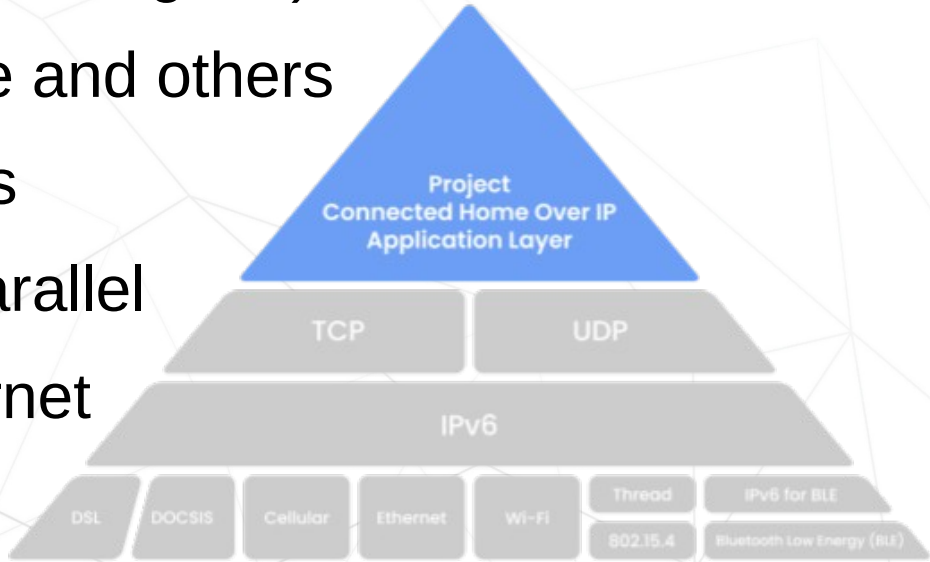
- Very active Open Source project (BSD3, but CLA)
- Border Router on Linux
- OpenThread nodes on Zephyr
- Product and app compatibility
- System plumbing for build, configuration and networking



Source: https://openthread.io/

Matter

- Connectivity Standards Alliance (former Zigbee)
- Driving companies are Google, Apple and others
- Final specification is work in progress
- SDK and test events developed in parallel
- Connectivity: WiFi, Thread and Ethernet
- IPv6 based



Source: https://github.com/project-chip/connectedhomeip/

Matter

- Active Open Source project (Apache 2)
- Fast moving target, hard to integrate currently
- Application layer, data model and device types
- Using Zigbee cluster library
- To much development discussions in members-only forums :-(

Source: https://github.com/project-chip/connectedhomeip/

IPv6-only Protocols

- 6lowpan paved the way for IPv6 on low-power standards e.g. ieee802154 in OpenThread
- IPv6 as default in new protocols like Matter as well
- We start to see IPv6-only IoT networks like we have seen IPv6-only data-centers before
- Some transition techniques could be re-used

IPv6

- Overview and interesting attributes for discussion
- Native IPv6 connectivity, prefix delegation
- NAT64 to transit through IPv4-only networks (e.g. your ISP)
- Tayga, jool or similar
- Connecting stub networks IETF draft
  https://datatracker.ietf.org/doc/html/draft-lemon-stub-networks-02

Updated IoT Devices

- Mandatory update functionality by law is coming in some countries
- ASOS approach: direct OTA on powerful devices and assisted-OTA on devices that could not spare the resources
- Updates: Protocol specific vs application specific vs platform specific
- ASOS will provide platform specific updates
- Application and protocol specific can come through containerized services

Zephyr & Linux

- On the ASOS side we build for both from Yocto
- Gateway would be Linux, but sensors most likely Zephyr
- Sharing of projects and libraries: mbedtls, openthread, matter

Thanks!

Appendix

ieee802154 mlme

- Advanced functionality like scanning, joining, network coordinator, etc needs support for command and beacon frame processing
- Over the years various parties started work but it never found its way into mainline
- To make linux-wpan an option for real-world deployments this will be needed

Native ieee802154 Transport

- OpenThread as border router runs natively on Linux
- It uses a radio co-processor or network co-processor architecture where Linux networking is only involved on top
- PHY, MAC, 6lowpan are all firmware or userspace driven
- Alex mentioned raw frames with AF_PACKET which would, again, circumvent parts of the kernel stack
- More functionality needed if we want a linux-wpan OpenThread platform abstraction